Fiscal Year:	FY 2007	**Task Last Updated:**	FY 07/20/2007
PI Name:	Albert, Mitchell S Ph.D.		
Project Title:	Hyperpolarized Noble Gas Magnetic Resonance Imaging		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01655	**Congressional District:**	3
Comments:	Note: PI moved to U Mass Medical School in 2007; previously at Brigham and Women's Hospital/Harvard University (9/2008)		
Project Type:	Ground	**Solicitation / Funding Source:**	01-OBPR-07
Start Date:	01/10/2003	**End Date:**	01/31/2007
No. of Post Docs:	3	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed back to 1/31/2007 per J. Dardano/JSC, as NCE referred to same project with new grant number (info rec'd 9/2008) NOTE: Received no-cost extension to 1/31/2008 per P. Denkins/JSC (3/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mulkern, Robert (Children's Hospital Boston) Reilly, John Joseph (Brigham and Women's Hospital)		
Grant/Contract No.:	NAG9-01469		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A novel biomedical imaging technique, hyperpolarized (HP) noble gas magnetic resonance imaging (MRI) is capable of generating high-resolution images of the lung, heart, and brain by using the spin-1/2 noble gases, 3He and 129Xe. The HP gases, created via spin-exchange optical pumping, have greatly enhanced MR detectability. Unlike conventional proton MRI, HP gas MRI does not require a large magnetic field and, therefore, can be performed in very low magnetic fields (~ 0.01 Tesla). Thus, a small, portable, low-power MRI system, practical for space-flight, may be designed to implement very low field HP gas MRI. The imaging capacity of HP gas MRI itself is also relevant to space flight research. Under prior NASA funding, we have obtained (i) cerebral blood flow measurements and (ii) lung gas transfer measurements using HP 129Xe MRI in rats. Additionally, in humans we have (i) obtained high-resolution images of the distal airways in lungs up to the 7th generation, (ii) created models of the three-dimensional lung airway tree, and (iii) obtained very low field (0.015 Tesla) images of the lungs, using HP 3He MRI. We now propose to apply this novel technique to investigations of the physiological and functional states of the lungs, heart, and brain. Studies will focus on regional ventilation and perfusion inhomogeneities of the lungs, patent foramen ovale and pulmonary embolism of the heart, and cerebral function and blood flow of the brain. Our proposed ground-based investigations with HP gas MRI will impact space flight research in three important ways: (i) by facilitating the acquisition of novel information about the physiological structure and function of organs and systems most affected by changes in microgravity; (ii) by developing a new technology that may have clinical applications for the diagnosis of pathologies endemic to microgravity environments; and (iii) by developing treatment strategies that address the continuum of a space-induced illness.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Today, more than ever, the understanding of pulmonary function has become critical to the treatment of disease and to the health of the community at large. Currently in the U.S., pulmonary disease affects over 20 million people, results in 123,000 deaths per year, and is the fourth leading cause of death. The economic cost of pulmonary disease is estimated to be over ten billion dollars a year. The personal cost, although not quantifiable, is no doubt enormous. The understanding of environmentally induced pulmonary illness has also taken on more importance with recent advances in farming and industry. Endotoxin exposure appears to be a growing cause of airway disease in workers in livestock and agricultural operations, while environmental exposure to asbestos is the major risk factor for development of mesothelioma (cancer of the lung pleura). And most recently, the threat of biological terrorism using exposure to inhaled toxins poses the potential for catastrophic morbidity and mortality. Given this vast landscape of costly threats to the pulmonary health of the U.S. population, the economic necessity of more accurately understanding, diagnosing, and preventing naturally occurring and environmentally induced pulmonary diseases cannot be understated. HP Noble Gas MRI is quickly moving to the forefront of clinical techniques that allow physicians to more quickly and accurately diagnose natural and environmentally induced pulmonary diseases. Rapid diagnosis and staging of disease with HP Noble Gas MRI will allow earlier therapeutic intervention and thus better health outcomes, all at a cost savings for individuals and the healthcare community. HP Noble Gas MRI will also offer cost savings by increasing the versatility, resolving power and even portability of existing imaging facilities, and thereby allow the more widespread application of MRI in major medical centers and its use in smaller clinics. Because the SNR of HP Noble Gas MRI is virtually independent of magnetic field strength, the technique does not require high magnetic fields, and can be performed with cheaper magnets. This allows HP Noble Gas MRI to be carried out in most any clinical setting that has an MRI machine. Such units could be used for low-cost, portable imaging and disease diagnosis in a variety of primary care facilities, especially those which specialize in the treatment of underserved persons, and which are adequately connected to social service infrastructure, handicapped accessible, public transportation, and other services designed to assist persons with disabling conditions (e.g., visiting nurse, home health aides, child and elder care). A portable imaging device would further benefit the very persons who, by virtue of their disabling condition, and/or inability to access quality healthcare, cannot demand the best or most optimal treatment. HP Gas MRI is also expected to have an impact of the quality and kind of research that can be carried out to study the etiology and treatment of pulmonary disease. HP 129Xe can be easily and non-invasively delivered to animals so that the same animal can be used over the course of a study. Not only does this alleviate the need for sacrificing a large number of animals, it more importantly allows the study of a disease process or experimental therapeutic treatment over time, which makes it directly relevant to the study of diseases and therapeutics in human clinical trails. We are convinced that HP Noble Gas technology will one day transform the way disease is detected, diagnosed, and treated, and we are specifically committed to developing assistive imaging technologies for a wide range of pulmonary and neurological disorders—both congenital and acquired. It is our firm hope and expectation that HP Noble Gas imaging will not only shed light on heretofore elusive disease mechanisms; it will exert a positive impact on patient outcomes, quality of life, and overall longevity.
Task Progress:	Experiments are ongoing for imaging the ventilation and airways of asthmatic and non-asthmatic subjects with HP 3He MRI. With simultaneous lung mechanics measurements, lung ventilation are being mapped and compared between the two subject populations, and the associated airway structure are being correlated with lung mechanical properties. In addition to our lung studies, advances have been made in our effort to develop hyperpolarized 129Xe MRI for the assessment of brain structure and function. The three most significant advances in this area include: 1. The method of chemical shift imaging was validated in our laboratory. 2. Hyperpolarized 129Xe MRI CSI was carried out in rat brain. 3. In vivo Functional Brain Imaging Using Hyperpolarized 129 Xe was demonstrated. Previously, we reported that the rigorous airway diameter quantification algorithm we developed has been tested on an anthropomorphic lung phantom, and presented the results. In the past year, 5 subjects who have had their airways scanned using HP 3He MRI have also been imaged with CT. The algorithm computing the local ventilation coefficient of variation from HP 3He MR images previously developed was applied to all ventilation data sets available. The resulting heterogeneity maps, viewed in the context of the ventilation maps, clearly demonstrate the capture of the ventilation heterogeneity characteristics. The algorithms for mapping ventilation have been applied to 4 non-asthmatics and 13 asthmatic subjects. Statistical significant differences were observed in the heterogeneity scores as well as ventilation distribution standard deviation

	for various cases of comparison. The results of this study support the idea that HP 3He MRI can serve as a diagnostic tool that can be regularly used to evaluate astronaut physiology, how that differs from their earth status, how it progresses with continued space flight, and how it alters in the presence of disease or impairment. [editor's note: Project continues with grant number NNX07AP82G; see that project for FY08 forward (9/08)]
Bibliography Type:	Description: (Last Updated: 03/23/2012)
Abstracts for Journals and Proceedings	Tzeng Y-S, Shah NM, Mansour JK, Gereige J, Washko GR, Cho MH, Stepp E, Lutchen KR, Albert MS. "Response to Deep Inspirations in Reactive and Non-Reactive Asthmatic Subjects Using Hyperpolarized 3He MRI." To be presented at the 15th Annual Meeting ISMRM, Berlin, Germany, May 2007. Proceedings of ISMRM 15th Annual Meeting and Exhibition, May 2007. , May-2007
Abstracts for Journals and Proceedings	Tzeng Y-S, Gereige J, Szender JB, Zhalehdoust-Sani S, Washko GR, Cho MH, Shah NM, Mansour JK, Stepp E, Albert MS, Lutchen KR. "Similar Responses to Deep Inspirations Observed in Reactive and Non-Reactive Asthmatic Subjects Using Hyperpolarized 3He MRI." To be presented at the American Thoracic Society Meeting 2007, San Francisco, May 2007. Proceedings of the American Thoracic Society 2007. , May-2007